Class Adm No

Candidate Name: ____________________

Class	Adm No

H

millennia institute

2018 Preliminary Exams
Pre-University 3

MATHEMATICS **9758/01**

Paper 1 **11 Sept 2018**

3 hours

Additional Materials: Answer Paper
List of Formulae (MF26)

READ THESE INSTRUCTIONS FIRST

Write your admission number, name and class on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams or graphs.
Do not use paper clips, highlighters, glue or correction fluid.

Answer **all** the questions.
Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.
You are expected to use an approved graphing calculator.
Unsupported answers from a graphing calculator are allowed unless a question specifically states otherwise.
Where unsupported answers from a graphing calculator are not allowed in a question, you are required to present the mathematical steps using mathematical notations and not calculator commands.
You are reminded of the need for clear presentation in your answers.

At the end of the examination, arrange your answers in NUMERICAL ORDER and fasten all your work securely together.

The number of marks is given in brackets [ ] at the end of each question or part question.

This question paper consists of 7 printed pages.

[Turn over

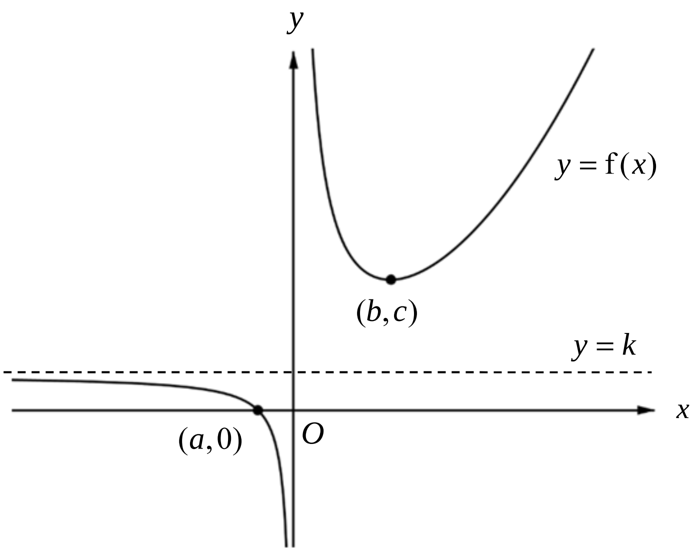
1 By using the series expansion of e^x from the List of Formulae (MF26), find $\sum_{r=2}^{\infty} \frac{3^r}{r!}$. [3]

2 Without using a calculator, solve the inequality

$$\frac{x-7}{x^2-x-6} \leq 1. \qquad [4]$$

Hence solve the inequality $\frac{\ln x - 7}{(\ln x)^2 - \ln x - 6} - 1 \leq 0$. [2]

3 The diagram shows a sketch of the graph of $y = f(x)$. The curve intersects the x-axis at the point $(a, 0)$ and has a turning point at the point (b, c), where a, b and c are constants. The y-axis and the line $y = k$, where k is a constant, are the asymptotes of the curve.



Sketch, on separate diagrams, the graphs of

(i) $y = \frac{1}{f(x)}$, [3]

(ii) $y = f'(x)$, [3]

stating clearly, where possible, the equations of any asymptotes and coordinates of any turning points and axial intercepts.

[Turn over

4 The curve C has equation $y=\dfrac{2x^2+13x+23}{x+\lambda}$, where λ is a positive constant.

(i) Sketch the graph of C, stating clearly the coordinates of any points of intersection with the axes and equations of any asymptotes. [3]

(ii) For $\lambda=3$, find algebraically the set of values that y can take. [3]

5 **(i)** Given that $\ln y=\sin 2x$, show that $\dfrac{\mathrm{d}y}{\mathrm{d}x}=2y\cos 2x$. Hence find the values of $\dfrac{\mathrm{d}y}{\mathrm{d}x}$ and $\dfrac{\mathrm{d}^2y}{\mathrm{d}x^2}$ when $x=0$. [3]

(ii) Write down the first three non-zero terms in the Maclaurin series for y. [1]

(iii) It is given that the second and third non-zero terms in part **(ii)** are equal to the first and second non-zero terms in the series expansion of $\mathrm{e}^{px}\ln(1+qx)$ respectively. Using appropriate expansions from the List of Formulae (MF26), find the values of the constants p and q. [3]

Hence state the range of values of x for which the series expansion of $\mathrm{e}^{px}\ln(1+qx)$ is valid. [1]

6 A curve C has parametric equations

$$x=1-\mathrm{e}^{-t},\qquad y=1+t^2.$$

(i) Find the exact gradient of C when $t=1$. [2]

(ii) By first finding an expression for $\dfrac{\mathrm{d}^2y}{\mathrm{d}x^2}$ in terms of the parameter t, determine if C is concave upward for all real values of t. [2]

(iii) Find the cartesian equation of C. [2]

(iv) Hence find the volume of revolution when the region bounded by C, the x-axis and the lines $x=-\dfrac{1}{2}$ and $x=\dfrac{3}{5}$ is rotated completely about the x-axis, giving your answer correct to 3 decimal places. [2]

7 **(a)** **(i)** Find $\int \cos 2x \sin^5 2x \, \mathrm{d}x$. [3]

(ii) Use the substitution $u = \mathrm{e}^x$ to find $\displaystyle\int \frac{\mathrm{e}^x}{\sqrt{1-2\mathrm{e}^{2x}}} \, \mathrm{d}x$. [3]

(b) **(i)** Show that $\displaystyle\int x^2 \sin nx \, \mathrm{d}x = \frac{2}{n^3}\cos nx + \frac{2x}{n^2}\sin nx - \frac{x^2}{n}\cos nx + c$, where n is a positive integer and c is an arbitrary constant. [3]

(ii) Hence find $\displaystyle\int_{\pi}^{2\pi} x^2 \sin nx \, \mathrm{d}x$, giving your answer in the form $\displaystyle\frac{p}{n^3} - \frac{q\pi^2}{n}$, where the possible values of p and q are to be determined. [3]

8 The function f is defined by

$$\mathrm{f} : x \mapsto \sqrt{3}\cos x - \sin x, \quad x \in \mathbb{R}, \ -\pi \le x \le \pi.$$

(i) Express $\mathrm{f}(x)$ as $R\cos(x+\alpha)$, where R and α are constants to be found. [1]

(ii) Sketch the graph of $y = \mathrm{f}(x)$, stating clearly the exact coordinates of the end-points and turning points. Hence explain why f^{-1} does not exist. [2]

(iii) If the domain of f is further restricted to $-\dfrac{\pi}{6} \le x \le k$, state the maximum value of k for which f^{-1} exists. [1]

(iv) Using the value of k found in part **(iii)**, find $\mathrm{f}^{-1}(x)$ and state the domain of f^{-1}. [3]

Hence sketch, on the same diagram, the graphs of $y = \mathrm{f}(x)$, $y = \mathrm{f}^{-1}(x)$ and $y = \mathrm{f}^{-1}\mathrm{f}(x)$. [3]

The function g is defined by

$$\mathrm{g}(x) = \frac{1}{1-x}, \quad x \in \mathbb{R}, \ x \ne 0, \ x \ne 1.$$

Given that $\mathrm{g}^2(x) = \dfrac{x-1}{x}$ and $\mathrm{g}^3(x) = x$ for $x \in \mathbb{R}$, $x \ne 0$, $x \ne 1$, determine the value of $\mathrm{g}^{2018}(2)$. [2]

9 A toy all-terrain vehicle, Terra is designed to handle multiple terrains. In a test run, the vehicle moves on a path that consists of different sections of increasing difficulty. Terra completes the first section, second section, third section and so on until all sections have been completed. The total time (in seconds) taken by Terra to complete *n* sections is given by $\frac{3}{2}n^2+\frac{13}{2}n$.

(i) Find the time taken by Terra to complete the *n*th section. [2]

Terra was made to travel up a man-made hill of height 35 m, from the foot of the hill to the peak. Terra is designed to cover a vertical distance of 5 m in the first ten minutes. For each subsequent ten-minute period, Terra covers 80% of the vertical distance covered in the previous ten-minute period.

(ii) The vertical distances covered by Terra in the ten-minute periods form a geometric sequence. Explain if this sequence is convergent. [1]

(iii) Explain if Terra is able to reach the peak of the hill. [2]

After the completion of the fourth ten-minute period, Terra's vertical distance covered is changed such that in each subsequent ten-minute period, Terra covers 90% of the vertical distance covered in the previous ten-minute period.

(iv) Find the minimum number of ten-minute periods required for Terra to reach the peak of the hill. [3]

After installing the brake system on Terra, the design team wants to test its effectiveness. In this test, Terra moves with increasing speed from a stationary position for 4 seconds before the brake system is engaged. Terra then has to stop moving in 2 seconds. The test is done many times and is repeated immediately after the previous one.

Terra's motion is modelled by the function given by

$$\mathrm{f}(t)=\begin{cases} \frac{3}{2}t & , \quad 0\le t\le 4, \\ \frac{3}{2}t^2-18t+54 & , \quad 4<t\le 6, \end{cases}$$

where t denotes the time (in seconds) after the start of the first test and $\mathrm{f}(t)$ denotes Terra's speed (in metres per second) at time *t*. The repetition of the test can be represented by the relationship given by $\mathrm{f}(t)=\mathrm{f}(t+6)$ for $t\ge 0$.

(v) Sketch the graph of $y=\mathrm{f}(t)$ for $0\le t\le 14$. [2]

(vi) The distance covered by Terra is given by the area under the graph of $y=\mathrm{f}(t)$. Find the distance travelled by Terra from $t=4$ to $t=11$. [2]

10 Relative to the origin O, the position vectors of the points A and B are $\begin{pmatrix} 0 \\ 5 \\ 15 \end{pmatrix}$ and $\begin{pmatrix} -5 \\ 7 \\ 12 \end{pmatrix}$ respectively. The line l_1 has equation $\mathbf{r} = \begin{pmatrix} -2 \\ 1 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} -1 \\ 2 \\ 3 \end{pmatrix}$, $\lambda \in \mathbb{R}$. The line l_2 is parallel to the vector $\begin{pmatrix} -1 \\ 6 \\ 12 \end{pmatrix}$ and passes through A.

(i) Given that l_1 and l_2 intersect at the point C, find $\overrightarrow{OC}$. [2]

The point B lies on l_1 and is the foot of the perpendicular from A to l_1.

(ii) Find the cartesian equation of the line of reflection of l_2 in the line l_1. [2]

The equations of two planes π_1 and π_2 are as follows:

$$\pi_1: \quad x + y - 2z = -1,$$
$$\pi_2: \quad ax + y + 3z = 5,$$

where a is a constant.

(iii) Find the coordinates of the foot of the perpendicular from A to π_1 and hence, find the perpendicular distance from A to π_1. [4]

(iv) Given that the angle between l_1 and π_2 is 30°, find the value(s) of a. [2]

(v) Given instead that the line of intersection of π_1 and π_2 has the equation $\mathbf{r} = \mathbf{v} + \beta \begin{pmatrix} 5 \\ -1 \\ 2 \end{pmatrix}$, $\beta \in \mathbb{R}$, where $\mathbf{v}$ is the position vector of the point V on the line of intersection, determine the value of a and a possible position vector of V. [3]

11 A game designer created a computer game that involves an alien race inhabiting and developing a planet.

(i) The population of the aliens (in thousands) on the planet at time n years is denoted by x_n. The population of the aliens changes according to the relationship that the game designer devised as follows:

$$x_n = x_{n-1} + \frac{13}{2} - n \text{ for } n \geq 1.$$

The initial population of the aliens on the planet is 8000. By considering $\sum_{r=1}^{n}(x_r - x_{r-1})$, find an expression for x_n in terms of n. [4]

(ii) The game designer believes that the relationship devised in part **(i)** is overly simplistic and decides to use another model. The population of the aliens (in thousands) at time t years is denoted by x. It is set that the growth rate of x is proportional to $50 - 2t - x$.

It is given that the initial population of the alien on the planet is 8000 and the population grows at a rate of 14000 per year initially.

Show that the growth rate of x at time t years can be modelled by the differential equation

$$\frac{dx}{dt} = \frac{1}{3}(50 - 2t - x). \qquad [2]$$

Using the substitution $u = 2t + x$, find x in terms of t. [5]

(iii) To add more variation into the game play, the game designer decides to use an alternative model that is given by

$$\frac{d^2x}{dt^2} = -\frac{100}{(t+1)^3}, \ t \geq 0,$$

where x is the population of the aliens (in thousands) at time t years.

It is given that the initial population of the aliens on the planet is 8000 and the population grows at a rate of $41\frac{2}{3}$ thousand per year initially.

Find x in terms of t. [3]

End of Paper